

Support Ticket Triage Prototype

Architecture and why. One LangChain create_agent running on LangGraph replaces the custom loop. Python/uv locks dependencies; Pydantic validates complete conversations, tool arguments and results. Terminal and synchronous FastAPI share one batch validator and triage function, preventing divergent routing. Two scoped read-only tools retrieve synthetic customer history and classic RAG passages from Docker PostgreSQL/pgvector. Exact cosine top-five search fits the small corpus; source/version/mock metadata travels with passages. Application-owned tool records and citation/policy checks stay outside model authority. No persistent memory, second agent or GraphRAG is introduced in this phase.

Failures and safeguards. Six model requests and eight real tool calls cap each ticket; GPT timeout is 30 seconds, DB connect/query timeout five seconds, with no automatic retries/correction and a separate 40-step graph limit. Complete call batches are authorized before execution. Foreign customers, unknown/malformed calls, provider/tool failures, exhausted budgets or unverifiable output produce explicit human-support fallback. Critical incidents take routing priority over billing; other billing needs human review. Auto-response requires low urgency, known history, supporting retrieved evidence and a draft; replies are never sent. Mock evidence and missing history remain disclosed. Thai drafts stay Thai. Prompt/tool data are untrusted; parameterized retrieval cannot execute arbitrary SQL. Citations prove source membership, not semantic truth. HTTP uses 422 for invalid input, sanitized 503 for unavailable config and 200 with per-ticket fallback for accepted runs; health is liveness only.

Evidence and production evaluation. Scripted chat models and mocked provider HTTP exercise the real graph, both tools, all twelve source messages, Docker retrieval, failures, CLI/API parity and overlapping request isolation. Fake lexical vectors prove contracts, not semantic relevance; live GPT/embedding quality remains unverified. Before production, evaluate held-out English/Thai/mixed tickets for urgency/action and extraction accuracy, critical false negatives, citation support, unsupported drafts, recall@5/MRR, prompt injection, latency, token cost and fallback rate. Repeat live runs, review failures with humans and add approved real knowledge, auth/tenant controls and observability before real customer use.